

CUSTOMER CHURN PREDICTION PROJECT REPORT

1. Introduction

Customer churn is a major challenge for subscription-based businesses, especially in the telecommunications sector. Retaining customers is significantly more cost-efficient than acquiring new ones, which makes churn prediction an essential analytical function. This project focuses on analyzing customer behavior, identifying churn patterns, and building a machine learning model to predict whether a customer is likely to leave the service.

2. Business Problem Statement

The organization experiences a steady loss of customers each month, impacting revenue and long-term growth. To reduce churn, the company needs a reliable prediction system and a deeper understanding of churn drivers.

The goal of this project is to:

- Identify customer segments with high churn risk.
 - Understand the factors that influence churn using detailed EDA.
 - Build a predictive model to classify customers as “Churn” or “Not Churn.”
 - Suggest actionable strategies to improve customer retention.
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3. Objectives of the Project

- Perform detailed exploratory data analysis to understand patterns.
 - Identify key factors influencing churn.
 - Build and evaluate a machine learning model.
 - Develop a user-friendly interface for churn prediction (Streamlit App).
 - Provide business recommendations based on findings.
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4. Dataset Overview

The dataset contains customer-level information including demographics, contract details, billing patterns, services subscribed, and churn status.

Feature Groups

- **Demographic:** Gender, Partner, Dependents, SeniorCitizen
 - **Services:** PhoneService, InternetService, Security, Backup, Streaming, Tech Support
 - **Account Info:** Contract Type, Payment Method, Paperless Billing
 - **Billing:** MonthlyCharges, TotalCharges
 - **Target Variable:** Churn (Yes/No)
 - **Total Records:** 7043
 - **Total Features:** 21
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5. Exploratory Data Analysis

This section provides an in-depth analysis of the customer dataset to identify patterns, trends, and relationships that influence churn.

5.1 Initial Observations

- MonthlyCharges range widely indicating diverse service packages.
 - Tenure varies from new customers (0 months) to highly loyal ones (72 months).
 - TotalCharges has zeros for new subscribers.
 - Churn rate is **approximately 26.5%**, showing significant customer loss.
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5.2 Summary Statistics

Numerical Feature Summary

Feature	Min	Max	Mean	Comments
Tenure	0	72	32.37	Highly variable – clear separation of new vs loyal customers
MonthlyCharges	18.25	118.75	64.76	Higher charges often linked to churn
TotalCharges	0	8684.80	2283.30	Zero values show first-month subscribers

Key Insights

- Significant portion of customers leave within the first few months.
- Higher billing customers tend to churn faster.
- Tenure is highly correlated with loyalty.

5.3 Distribution of Target Variable

Described Visual:

“A bar chart shows that approximately one-fourth of customers have churned. This imbalance indicates a need for careful model evaluation.”

5.4 Univariate Analysis (Feature-by-Feature)

Tenure

“A histogram reveals a large concentration of customers in the first 0–6 months, indicating onboarding issues.”

Monthly Charges

“Density plots show two major groups: low billers (\$20–\$45) and high billers (\$70–\$100). The higher billing group has a higher churn rate.”

Total Charges

“Customers with lower TotalCharges have higher churn rates, especially those below \$500.”

5.5 Bivariate Analysis (Churn vs Features)

Churn vs Gender

“Both genders show almost identical churn rates, indicating gender does not influence churn.”

Churn vs Senior Citizen

“Bar charts show senior citizens have significantly higher churn, possibly due to pricing issues or service complexity.”

Churn vs Partner & Dependents

“Customers with families (partners/dependents) churn less.”

Churn vs Contract Type

“One of the clearest patterns:

- Month-to-Month → Highest churn
- One-Year → Moderate
- Two-Year → Lowest”

Churn vs Payment Method

“Electronic check users show very high churn, while automatic payments (credit card/bank transfer) reduce churn.”

Churn vs Paperless Billing

“Paperless billing users churn more, strongly linked with flexible contracts.”

Churn vs Internet Service

“Fiber optic users churn more, suggesting dissatisfaction with network reliability.”

Churn vs Security and Tech Services

“For all services (TechSupport, OnlineSecurity, Backup, Protection), customers who opt in show reduced churn.”

5.6 Multivariate Analysis Correlation

Heatmap (Described)

“A heatmap showed:

- Strong positive correlation: Tenure ↔ TotalCharges
- Weak but positive: MonthlyCharges ↔ Churn
- Negative: Tenure ↔ Churn
- Contract influences churn indirectly through tenure”

Monthly Charges + Tenure Interaction

“3D scatter plot shows customers with high charges AND low tenure churn the most.”

Contract + Billing + Churn

“Visuals confirm: high monthly charges + month-to-month contract = churn hotspot.”

5.7 Hidden Patterns Identified

- New customers are most vulnerable to churn.
- Fiber optic customers churn more despite being premium service users.
- Add-on services (Tech, Security) significantly reduce churn.
- Families (partners/dependents) show more loyalty.

- High billing with no long-term contract = highest churn group.
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6. Data Preprocessing

- Converted TotalCharges to numeric
 - Handled blank values
 - Encoded categorical features
 - Standardized inputs
 - Split data into train-test sets
 - Saved model and encoders for deployment
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7. Machine Learning Model

Algorithms Tested

- Logistic Regression
- Random Forest Classifier
- XG Boosting

Best Performer: Random Forest

Model Results (Text Format)

- **Accuracy:** 82.4%
- **Precision (Churn Class):** 0.69
- **Recall (Churn Class):** 0.71
- **F1 Score:** 0.70

“Model shows strong recall, meaning it catches most churners — useful for retention strategies.”

8. Feature Importance

Described Visual:

“A bar plot ranked Contract, Tenure, MonthlyCharges, OnlineSecurity, and TechSupport as the most important predictors.”

9. Streamlit App

The application includes:

Input Form Section

- Dropdowns for categorical features
- Sliders for numerical values
- Clean layout for entering customer details

Prediction Section

- Displays churn probability
- Shows whether customer is “Likely to Churn”

Visualization Section

Described visuals included:

- Tenure distribution chart
- Churn by contract type
- Billing analysis

10. Business Recommendations

- Improve early customer onboarding to reduce first-month churn.
 - Encourage long-term contracts with incentives.
 - Review fiber optic service performance.
 - Promote security/tech support add-ons.
 - Target high billing customers with discounts.
 - Introduce loyalty benefits to reward tenure.
 - Focus on customers who use electronic checks — introduce easier payment options.
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